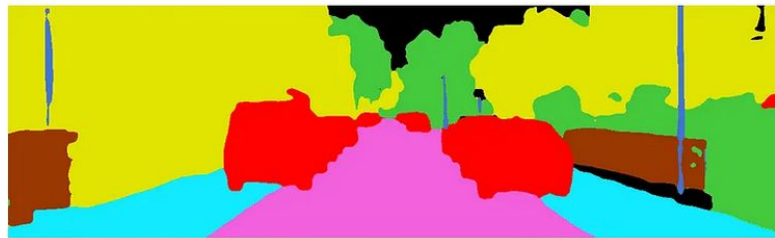

Semantic Segmentation of Indoor Scenes

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Project Overview

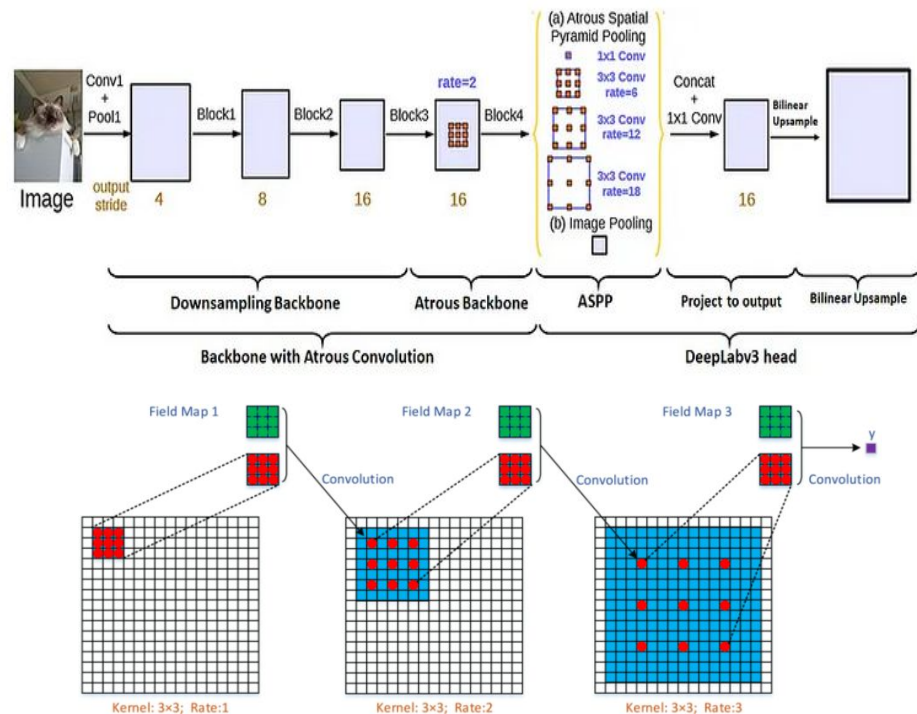
- Semantic Segmentation – field of computer vision tasked with categorizing pixels in images
- Task – Given an image, output a colored mask with predicted class for each pixel
- This project is a system to dynamically segment indoor scenes in real time given a camera input



 Road	 Sidewalk	 Building	 Fence
 Pole	 Vegetation	 Vehicle	 Unlabel

Model Architecture

- Uses DeepLabV3 model architecture as a starting point
 - Features a ResNet50 convolutional backbone which identifies features in the image
 - Atrous Spatial Pyramid Pooling (ASPP) layer works by inserting spaces between fields, allowing for convolution to capture different levels of information



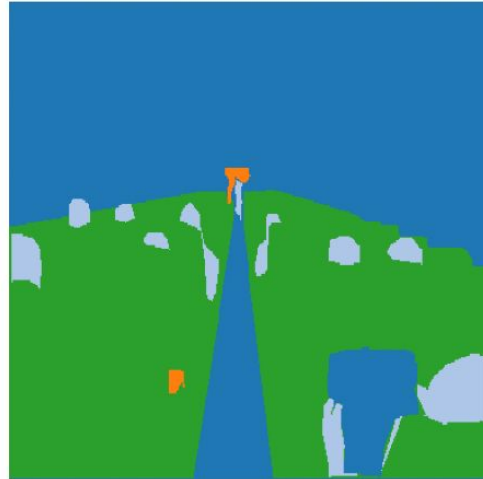
Dataset

- Data for semantic segmentation requires pixel-level annotations for each image, making large captured datasets impractical
- Dataset Contents – Our captured images/labeled masks combined with data from the ADE20K dataset
 - ADE20K dataset contains over 27K image/mask pairs, with 150 classes
- Augmentation introduces minor visual changes (blurring, deformations) to improve training outcomes

Image



Ground Truth



Training and Results

- Google Colab was used for training, due to the high computational requirement
- Trained for ~200 epochs until accuracy gains started to slow down
- Mean Intersection over Union (mIoU) score used to evaluate performance
- Final Results:
 - Train mIoU: 0.751
 - Validation mIoU: 0.613
- After model was finalized, we converted it into a TensorRT engine to use for live inference on the Jetson Orin Nano

Live Demo

Any questions are welcome!